

Advanced Programming Practices

MTech 2026

Lab - 9

General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo_APP9_Qn.c (e.g. ABC2026001_APP9_Q1.c).
 - Every program must compile without warnings using the gcc compiler.
 - Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
 - Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
 - Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.
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1. Write a menu driven program for performing stack operations such a PUSH, POP, finding top of the stack.
2. Implement stack operations as in program 1 using singly linked list instead of array.
3. Write a C Program to implement Queue using an array. The program should be menu-driven, with the option to enqueue, and dequeue, to check whether the queue is empty or full, traverse, and exit based on the user's choice.
4. Write a C Program to implement Circular Queue using an array. The program should be menu-driven, with the option to enqueue, and dequeue, to check whether the queue is empty or full, traverse, and exit based on the user's choice.
5. Implement Circular Queue and perform operations on it using doubly circular linked list instead of array.